PARLAPALLI MANIKANTESWARA REDDY

Data Scientist | AI/ML & Data Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate with a strong foundation in statistics, mathematics, and computer science, and hands-on experience in Python, SQL, machine learning, and deep learning. Skilled in mining and analyzing data to identify trends and patterns, developing predictive models and algorithms using TensorFlow, Keras, and Scikit-learn, and performing exploratory data analysis to turn data into actionable insights. Experienced in Generative AI (LLMs, RAG, Prompt Engineering), REST API development, and cloud platforms, with an Oracle Cloud Infrastructure Generative AI Professional certification. Strong analytical, communication, and presentation skills.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, Object-Oriented Programming (OOP)

Machine Learning & Predictive Modeling: Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Predictive Models & Algorithms, TensorFlow, Keras, Scikit-learn

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, AI APIs

Data Mining & Analytics: Data Mining, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization, ETL

Frameworks & Tools: Pandas, NumPy, Matplotlib, Streamlit, FastAPI, Power BI, Excel

Databases: MySQL, Supabase

Cloud & Certifications: Oracle Cloud Infrastructure (OCI), Multi-Cloud Networking (Aviatrix)

PROJECTS

Landslide Susceptibility Prediction Using Machine Learning

- Developed a predictive model and algorithm to assess landslide susceptibility using environmental and terrain-related data, translating a real-world business/safety problem into a technical solution.
- Mined and analyzed multi-source terrain data to identify trends and patterns; performed data preprocessing and feature engineering using Python, Scikit-learn, Pandas, and NumPy.
- Built an interactive Streamlit dashboard for real-time risk prediction and data visualization, communicating findings through an accessible reporting interface.

Student Management System

- Designed and implemented a full-stack data application using FastAPI, Streamlit, and Supabase (cloud database) for efficient data management.
- Developed RESTful APIs with Pydantic-based data validation and integrated a cloud database to perform secure CRUD operations.
- Deployed the application on Render and Streamlit Cloud, demonstrating experience with cloud-based deployment environments.

INTERNSHIP EXPERIENCE

Data Science Trainee *10000 Coders*

- Completed hands-on training in data science, machine learning, and data analytics using Python-based tools and libraries.
- Performed data mining, exploratory data analysis (EDA), data visualization, and predictive model development using Python, Pandas, NumPy, and Scikit-learn.
- Used Power BI and Excel to analyze data and communicate findings and recommendations through reports and presentations.

EDUCATION

B.Tech, Computer Science and Engineering — SIMATS Engineering, Chennai *2021 – 2025*

CGPA: 8.5 / 10

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Cloud Infrastructure 2024 Generative AI Professional (1Z0-1127-24)
- Aviatrix Certified Engineer – Multi-Cloud Network Associate
- Introduction to Internet of Things (NPTEL)
- Participated in coding contests and data analytics competitions; volunteered in organizing college workshops and technology awareness programs.
- Engaged in public speaking and group discussions to strengthen communication skills.